Chapter 5 — System Design and Architecture with AI

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> **System design interviews are the most important hour of the loop, and AI changes them less than you think.** This chapter teaches you to argue *with* the AI in the room, not *for* or *against* it.

What you will learn

- The **architecture decisions AI cannot make for you** — and why interviewers test for them.
- A **fully worked walkthrough** of the most common system-design prompt ("Design Twitter / Uber / a fantasy app") with explicit AI integration.
- The **trade-off conversation** — how to discuss design choices in front of a panel using vocabulary that signals seniority.

By the end you will be able to walk into a system-design interview, ask "may I think out loud and use an LLM as a sounding board?", and either way handle the hour with confidence.

Lessons

1. **01-what-humans-still-own.md** — Six categories of architecture decision where AI is a sounding board at best. Each category is a question your interviewer is checking you can answer.
2. **02-design-with-ai-walkthrough.md** — A 60-minute system-design walkthrough ("Design a fantasy-sports app for 1M users") with the AI integrated at every appropriate step.

3. **03-trade-off-conversations.md** — The vocabulary, structure, and pitfalls of "we have three options; here are the trade-offs" conversations.

How long it should take

- 45–60 min reading per lesson.
- 2–3 hours practicing one system-design problem with the AI integration.
- ~6 hours total.

Why this chapter is different

Chapters 3 and 4 were about producing code. This chapter is about producing *decisions*. AI accelerates code dramatically; it accelerates decisions only modestly. **Most of what an interviewer is looking for in a system-design hour is unchanged from 2020 — but the unchanged things are now more visible because the typing is no longer a distraction.**

If you're preparing for a senior or staff interview, this chapter is the one that tips the loop in your favor.

Before you start

You should have:

- Some familiarity with classical system-design topics (load balancing, caching, database sharding, queues, CDNs). If not, a 1-day primer like *System Design Interview* by Alex Xu or the educative.io system-design course will get you the vocabulary.
- Completed Chapter 2 (the seven pillars). The architecture chapter applies the pillars to design.
- A whiteboard or blank document for sketching diagrams as you read.

Open **01-what-humans-still-own.md**.